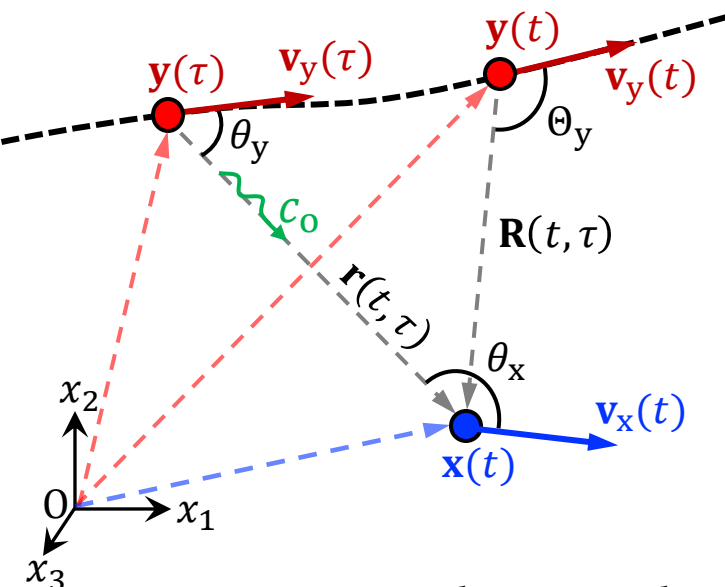


# Moving sources/observers

## Assumptions:

- **Air at rest** –  $\mathbf{v}_o = 0$
- **Causality** - for each emission time  $\tau$ , there is a unique reception time  $t > \tau$  along  $\mathbf{r}$  (no multi-valued mapping from  $\tau$  to  $t$  for along the path  $\mathbf{r}$ )
  - $\tau$  – sound source emission time – **retarded time**
  - $t$  – observer time when hearing the sound emitted at time  $\tau$
  - $\hat{\mathbf{n}}_r$  – line-of-sight unit vector from **source** (at emission time  $\tau$ ) to **observer** (at reception  $t$ )
- **Smooth motion** – source/observer motions are smooth enough to be differentiable
  - $\mathbf{y}$  – source position vector;  $\mathbf{v}_y$  – source velocity vector
  - $\mathbf{x}$  – observer position vector;  $\mathbf{v}_x$  – observer velocity vector
- **Subsonic observer motion** - observer does not move *exactly* at  $c_o$  along the line-of-sight so wavefronts never “catch” it



$c_o$  – sound wave speed

$$\text{Line-of-sight unit vector } \hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

❑ A wavefront is emitted by the source at time  $\tau$ , from position  $\mathbf{y}(\tau)$ , and is received by the observer at time  $t$ , at position  $\mathbf{x}(t)$

❑ We can define the distance vector from the **source** to **observer** as:  $\mathbf{r}(t, \tau) = \mathbf{x}(t) - \mathbf{y}(\tau)$

❑ Since sound propagates at speed  $c_o$ , we can write the following:  $|\mathbf{r}(t, \tau)| = c_o \cdot (t - \tau) \equiv |\mathbf{x}(t) - \mathbf{y}(\tau)|$

❑ **Retarded time:**

$$\tau = t - |\mathbf{x}(t) - \mathbf{y}(\tau)|/c_o$$

**Causality condition:**  $t > \tau$

# Moving sources/observers

- We shall now introduce the following function:

$$g(t, \tau) = t - \tau - \frac{|\mathbf{x}(t) - \mathbf{y}(\tau)|}{c_0} \equiv 0$$

- Based on causality ( $t > \tau$ ), we consider that  $t$  depends on  $\tau$ ,  $t = t(\tau)$

➤ A change in the emission time causes the reception time to change too

- We recognize that  $g(t(\tau), \tau) = 0$ , and assuming:

➤  $g(t, \tau)$  is a smooth function of  $t$  and  $\tau$ , satisfying  $g(t_0, \tau_0) = 0$

➤  $\partial g / \partial t \neq 0$  at any point  $(t_0, \tau_0)$

▪ Observer is not moving at  $c_0$  ...

- Computing the time derivative:

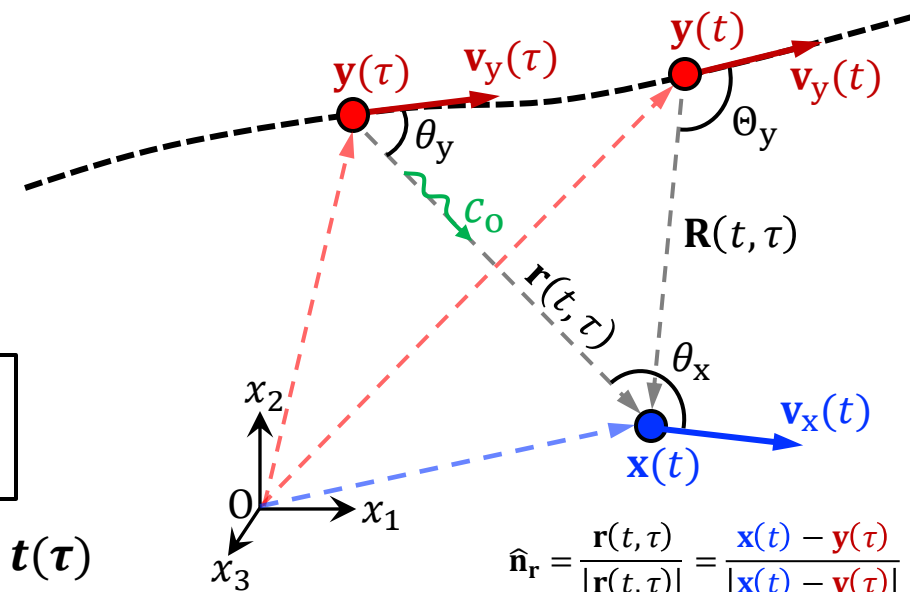
$$\frac{\partial g}{\partial t} = 1 - \frac{1}{c_0} \frac{\partial}{\partial t} |\mathbf{x}(t) - \mathbf{y}(\tau)| \Rightarrow \frac{\partial g}{\partial t} = 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \underbrace{\frac{\partial}{\partial t} (\mathbf{x}(t) - \mathbf{y}(\tau))}_{= \mathbf{v}_x(t)} \Rightarrow$$

$$\frac{\partial g}{\partial t} = 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_x(t) = 1 + M_x(t) \cos \theta_x$$

$\hat{\mathbf{n}}_r \cdot \mathbf{v}_x(t)$  – observer velocity in the source direction

$M_x(t) = \frac{|\mathbf{v}_x(t)|}{c_0}$  – observer Mach number

$\theta_x$  – angle between  $\mathbf{v}_x$  and  $\hat{\mathbf{n}}_r$



$$\hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \mathbf{r}(t, \tau)$$

$$\frac{d}{d\tau} [g(t(\tau), \tau)] = \frac{\partial g}{\partial t} \frac{dt}{d\tau} + \frac{\partial g}{\partial \tau} = 0$$

# Moving sources/observers

□ Similarly, the retarded time derivative is:

$$\frac{\partial g}{\partial \tau} = -1 - \frac{1}{c_0} \frac{\partial}{\partial \tau} |\mathbf{x}(t) - \mathbf{y}(\tau)| \Rightarrow \frac{\partial g}{\partial \tau} = 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \underbrace{\frac{\partial}{\partial \tau} (\mathbf{x}(t) - \mathbf{y}(\tau))}_{= -\mathbf{v}_y(\tau)}$$

$$\Rightarrow \boxed{\frac{\partial g}{\partial \tau} = -1 + \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_y(\tau) = 1 - M_y(\tau) \cos \theta_y}$$

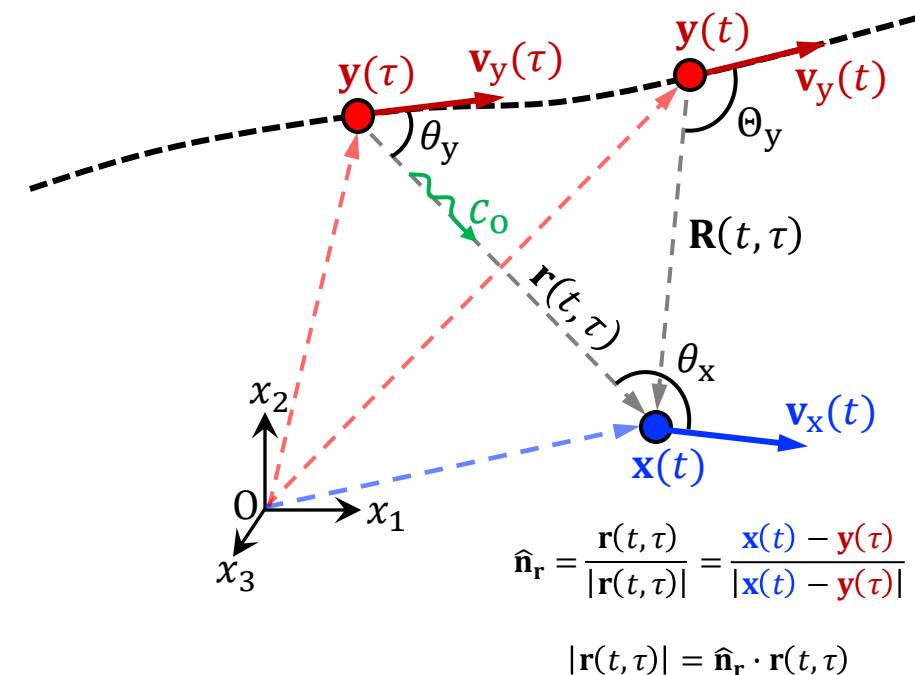
$\hat{\mathbf{n}}_r \cdot \mathbf{v}_y(\tau)$  – source velocity in the observer's direction

$M_y(\tau) = \frac{|\mathbf{v}_y(\tau)|}{c_0}$  – source Mach number

$\theta_y$  – angle between  $\mathbf{v}_y$  and  $\hat{\mathbf{n}}_r$

□ Substituting the above into  $dg/d\tau$  yields:

$$\Rightarrow \boxed{\frac{dt}{d\tau} = \frac{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_y(\tau)}{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_x(t)} = \frac{1 - M_y(\tau) \cos \theta_y}{1 + M_x(t) \cos \theta_x}}$$



$$\frac{dg}{d\tau} = \frac{\partial g}{\partial t} \frac{dt}{d\tau} + \frac{\partial g}{\partial \tau} = 0 \Rightarrow \left( 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_x(t) \right) \frac{dt}{d\tau} = - \left( -1 + \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_y(\tau) \right)$$

## Fundamental Kinematic Relation!

A small change in emission time  $d\tau$  leads to a corresponding change in reception time  $dt$

# Moving sources/observers

□ Assume a periodic sound signal is emitted with:

- $T_y = d\tau$  – emission time period
- $f_y = 1/d\tau$  – emitted frequency

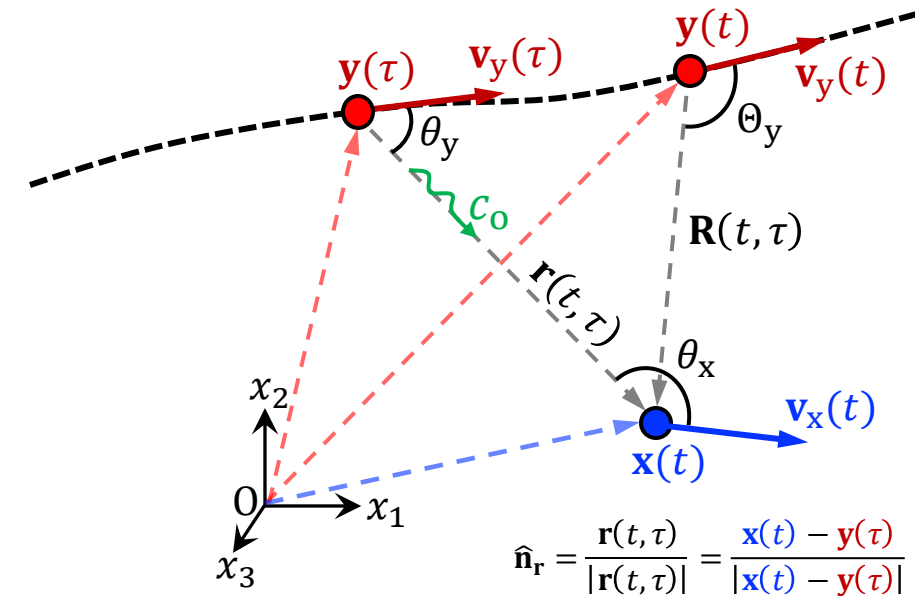
□ The observer received the sound signal with:

- $T_x = dt$  – reception time period
- $f_x = 1/dt$  – received frequency

□ Therefore, we obtain the **Doppler frequency shift equation**:

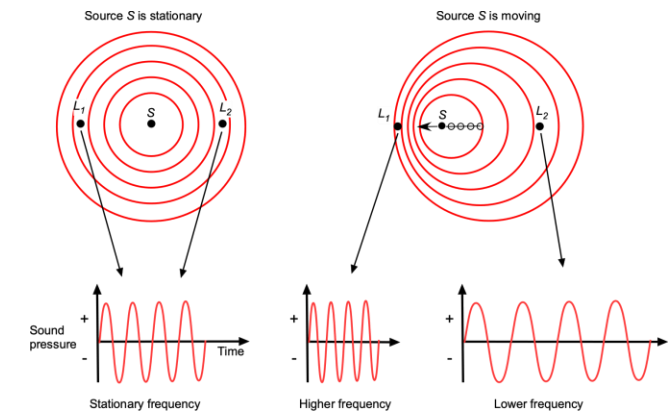
$$\frac{f_x}{f_y} = \frac{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_x(t)}{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_y(\tau)} = \frac{c_0 + |\mathbf{v}_x(t)| \cos \theta_x}{c_0 - |\mathbf{v}_y(\tau)| \cos \theta_y} = \frac{1 + M_x(t) \cos \theta_x}{1 - M_y(\tau) \cos \theta_y}$$

- $|\mathbf{v}_x(t)| \cos \theta_x < 0 \rightarrow$  *observer* moves **away** from the *source*
- $|\mathbf{v}_y(\tau)| \cos \theta_y < 0 \rightarrow$  *source* moves **away** from the *observer*



$$\hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \mathbf{r}(t, \tau)$$



# Moving sources/observers w. flow

- ❑ Consider now a **fluid medium with a uniform velocity  $\mathbf{v}_0$**
- ❑ In this scenario, we shall consider two frames of reference:
  - **Stationary 'observer' frame  $(\mathbf{x}, \mathbf{y}, t, \tau)$** 
    - Fluid medium moves with constant **velocity  $\mathbf{v}_0$**
  - **Moving fluid medium frame  $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{t}, \tilde{\tau})$** 
    - Fluid medium is at rest ( $\tilde{\mathbf{v}}_0 = 0$ )
- ❑ The Galilean transformation between the two frames is:

$$\begin{aligned}\tilde{\mathbf{x}}(t) &= \mathbf{x}(t) - \mathbf{v}_0 t \\ \tilde{\mathbf{y}}(\tau) &= \mathbf{y}(\tau) - \mathbf{v}_0 \tau\end{aligned}$$

- ❑ In the *moving frame* ( $\sim$ ) we can write:

$$\frac{d\tilde{\mathbf{x}}}{dt} = \frac{d}{dt}[\mathbf{x}(t) - \mathbf{v}_0 t] = \frac{d\mathbf{x}}{dt} - \mathbf{v}_0$$

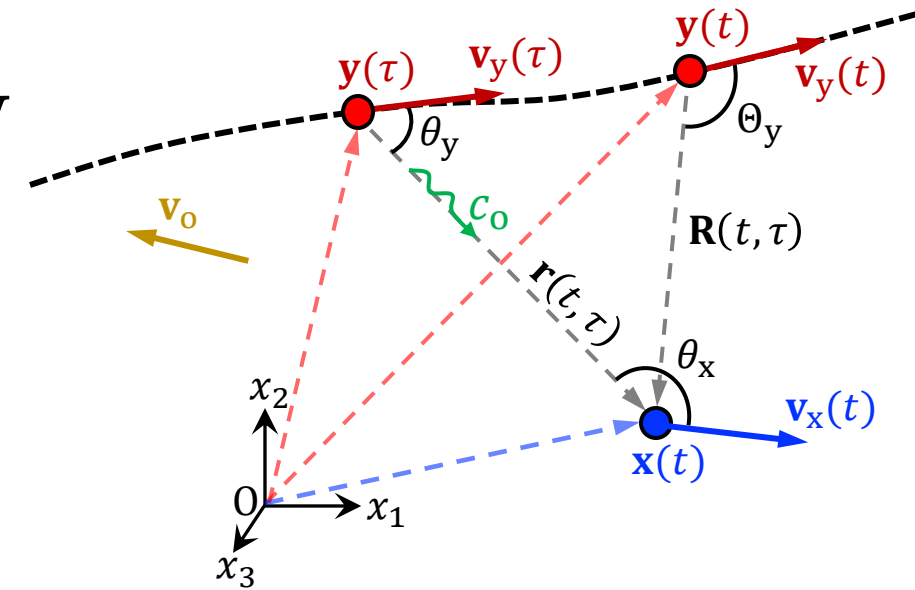
$$\frac{d\tilde{\mathbf{y}}}{d\tau} = \frac{d}{d\tau}[\mathbf{y}(\tau) - \mathbf{v}_0 \tau] = \frac{d\mathbf{y}}{d\tau} - \mathbf{v}_0$$



$$\begin{aligned}\tilde{\mathbf{v}}_x(t) &= \mathbf{v}_x(t) - \mathbf{v}_0 \\ \tilde{\mathbf{v}}_y(\tau) &= \mathbf{v}_y(\tau) - \mathbf{v}_0\end{aligned}$$

$$\begin{cases} t = \tilde{t} \\ \tau = \tilde{\tau} \\ f_x = \tilde{f}_x \\ f_y = \tilde{f}_y \end{cases}$$

Time is  
Galilean  
invariant...



$$\hat{\mathbf{n}}_r = \frac{\tilde{\mathbf{r}}(t, \tau)}{|\tilde{\mathbf{r}}(t, \tau)|} = \frac{\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)}{|\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_0(t - \tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_0(t - \tau)|}$$

$$|\tilde{\mathbf{r}}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \tilde{\mathbf{r}}(t, \tau)$$

$$\hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \mathbf{r}(t, \tau)$$

# Moving sources/observers w. flow

- We repeat the same analysis we previously did for fluid at rest, but now we perform it in the 'new' *moving frame* to obtain:

$$|\mathbf{r}(t, \tau)| = c_0 \cdot (t - \tau) \equiv |\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)| \xrightarrow{\text{define } g(t, \tau)=0} \frac{dt}{d\tau} = \frac{c_0 - \tilde{\mathbf{n}}_{\mathbf{r}} \cdot \tilde{\mathbf{v}}_{\mathbf{y}}(\tau)}{c_0 - \tilde{\mathbf{n}}_{\mathbf{r}} \cdot \tilde{\mathbf{v}}_{\mathbf{x}}(t)}$$

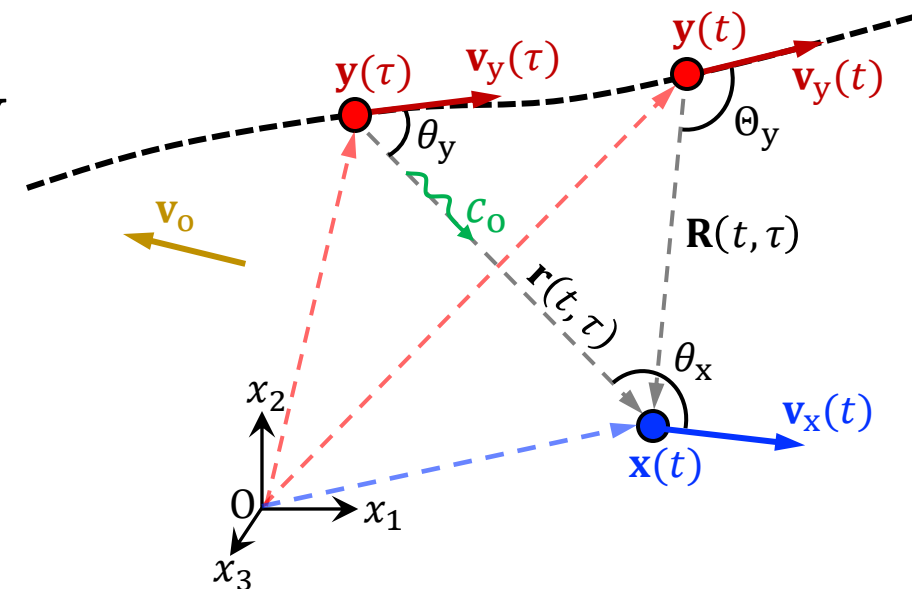
- $\tilde{\mathbf{n}}_{\mathbf{r}}$  – line-of-sight unit vector from **source** ( $\tau$ ) to **observer** ( $t$ ) in the *moving frame*

- Substituting back the stationary frame variables yields:

$$\begin{aligned} \tilde{\mathbf{v}}_{\mathbf{x}}(t) &= \mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0 \\ \tilde{\mathbf{v}}_{\mathbf{y}}(\tau) &= \mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0 \end{aligned} \Rightarrow \boxed{\frac{f_{\mathbf{x}}}{f_{\mathbf{y}}} = \frac{c_0 - \tilde{\mathbf{n}}_{\mathbf{r}} \cdot (\mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0)}{c_0 - \tilde{\mathbf{n}}_{\mathbf{r}} \cdot (\mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0)}}$$

- For most **low Mach** ( $M \ll 1$ ) uniform flows & **far-field observers** ( $|\mathbf{r}(t, \tau)| \gg 1$ ) we can approximate:

$$\tilde{\mathbf{n}}_{\mathbf{r}} \approx \hat{\mathbf{n}}_{\mathbf{r}} \Rightarrow \boxed{\frac{f_{\mathbf{x}}}{f_{\mathbf{y}}} \approx \frac{c_0 + |\mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0| \cos \tilde{\theta}_{\mathbf{x}}}{c_0 - |\mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0| \cos \tilde{\theta}_{\mathbf{y}}} = \frac{1 + M_{\mathbf{x}_r}(t) \cos \tilde{\theta}_{\mathbf{x}}}{1 - M_{\mathbf{y}_r}(\tau) \cos \tilde{\theta}_{\mathbf{y}}}}$$



$$\tilde{\mathbf{n}}_{\mathbf{r}} = \frac{\tilde{\mathbf{r}}(t, \tau)}{|\tilde{\mathbf{r}}(t, \tau)|} = \frac{\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)}{|\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_0(t - \tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_0(t - \tau)|}$$

$$|\tilde{\mathbf{r}}(t, \tau)| = \tilde{\mathbf{n}}_{\mathbf{r}} \cdot \tilde{\mathbf{r}}(t, \tau)$$

$$\hat{\mathbf{n}}_{\mathbf{r}} = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_{\mathbf{r}} \cdot \mathbf{r}(t, \tau)$$

$$M_{\mathbf{y}_r}(\tau) = \frac{|\mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0|}{c_0} \rightarrow \text{source Mach number relative to the flow}$$

$$M_{\mathbf{x}_r}(t) = \frac{|\mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0|}{c_0} \rightarrow \text{observer Mach number relative to the flow}$$

$$\tilde{\theta}_{\mathbf{y}} - \text{angle between } \mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0 \text{ and } \hat{\mathbf{n}}_{\mathbf{r}}$$

$$\tilde{\theta}_{\mathbf{x}} - \text{angle between } \mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0 \text{ and } \hat{\mathbf{n}}_{\mathbf{r}}$$

# Elementary acoustic sources

□ Elementary acoustic sources can be solved in two different ways:

➤ **Inhomogeneous wave equation**

- Sources are introduced *locally* either in the form of a mass flow  $q$  or in the form of a volume force  $\mathbf{F}$
- Involves **Green's functions** – will be discussed next...

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = \frac{\partial q}{\partial t} - \frac{\partial F_i}{\partial x_i} = Q(\mathbf{x}, t)$$

or

$$\frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = -i\omega q - \frac{\partial F_i}{\partial x_i} = Q(\mathbf{x}, \omega)$$

➤ **Homogeneous wave equation**

- Solution is achieved by setting boundary conditions to determine the unknown constants introduced by the sources
- We dealt with...

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = 0$$

or

$$\frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = 0$$

# Green's function

- A powerful mathematical tool to solve boundary value problems

What Green's function tells us?

Impulse response of an inhomogeneous linear differential operator (LDO)  $\mathcal{L}$  defined on a domain with specified initial conditions or BC



George Green first developed the concept in 1830s

- Suppose we have a linear ODE:

$$\mathcal{L}u(x) = f(x)$$

➤  $\mathcal{L}$  – differential operator

- A Green's function,  $G(x, s)$ , of a LDO  $\mathcal{L} = \mathcal{L}(x)$  acting on distributions over a subset of the Euclidean space  $\mathbb{R}^n$ , at a point  $s$ , is any solution of:

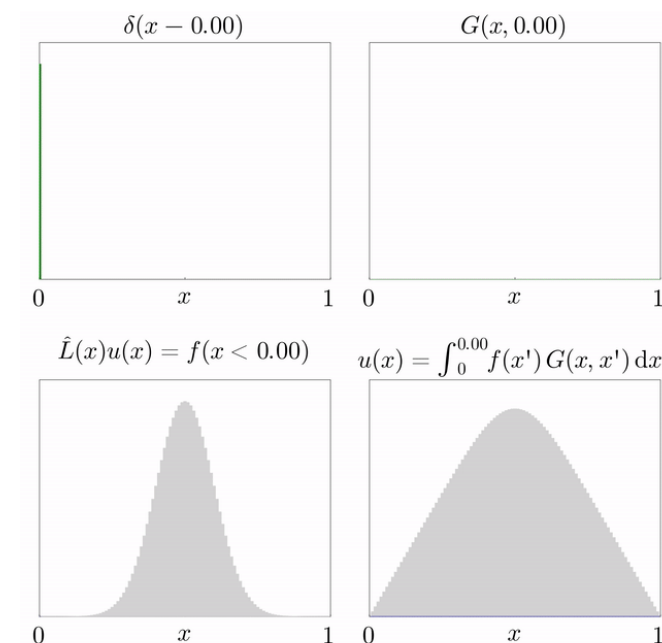
$$\mathcal{L}G(x, s) = \delta(x - s)$$

- If we know the linear operator response to a *point impulsive source*, how do we get the field for *any distribution of sources*? **Convolution!**

Dirac delta function

$$\delta(t - \tau) = \begin{cases} \infty, & \text{if } t = \tau, \\ 0, & \text{if } t \neq \tau. \end{cases}$$

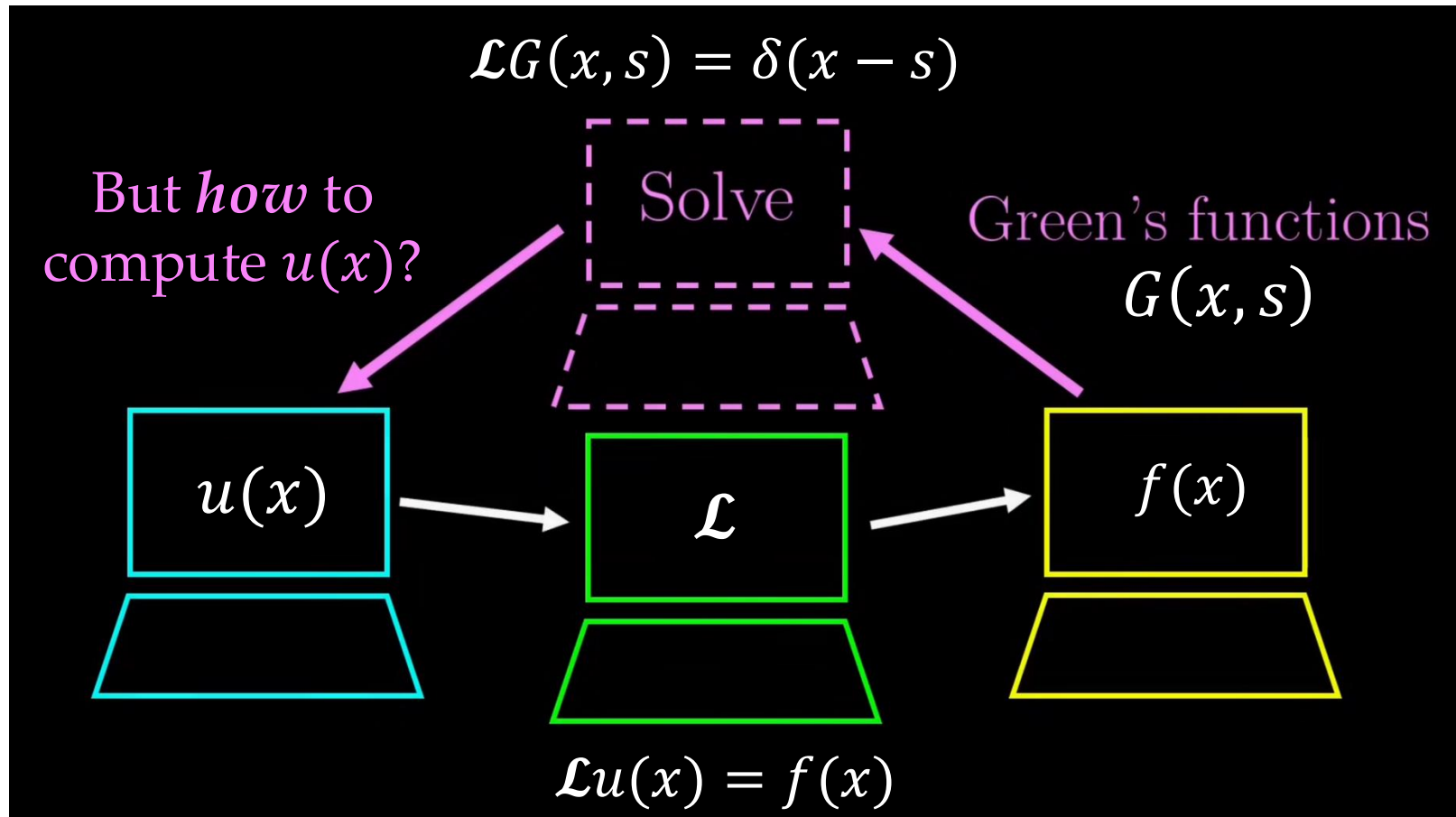
such that:  $\int_{-\infty}^{\infty} \delta(t - \tau) d\tau = 1, \quad -\infty < t < \infty$





# Green's function

Green's function  $G(x, s)$  provides us an *alternative* to compute  $u(x)$



# Green's function

$$\delta(t - \tau) = \begin{cases} \infty, & \text{if } t = \tau, \\ 0, & \text{if } t \neq \tau. \end{cases}$$

Sifting property

$$\int_{-\infty}^{\infty} f(t) \delta(t - \tau) dt = f(\tau)$$

- Assuming a Green's function  $G$  can be found for the operator  $\mathcal{L}$ , then:

$$\mathcal{L}G(x, s) = \delta(x - s) \xrightarrow[\text{and integrating over } s]{\text{Multiplying by } f(s)} \int \mathcal{L}G(x, s) f(s) ds = \int \delta(x - s) f(s) ds = f(x)$$

- Since  $\mathcal{L} = \mathcal{L}(x)$  is linear and acts only on the variable  $x$  (not on the integration variable  $s$ ), we may take  $\mathcal{L}$  outside of the integration:  $\Rightarrow \mathcal{L} \left( \int G(x, s) f(s) ds \right) = f(x)$

- Thus:

$$u(x) = \int f(s) G(x, s) ds$$

*convolution* ( $f * G$ )

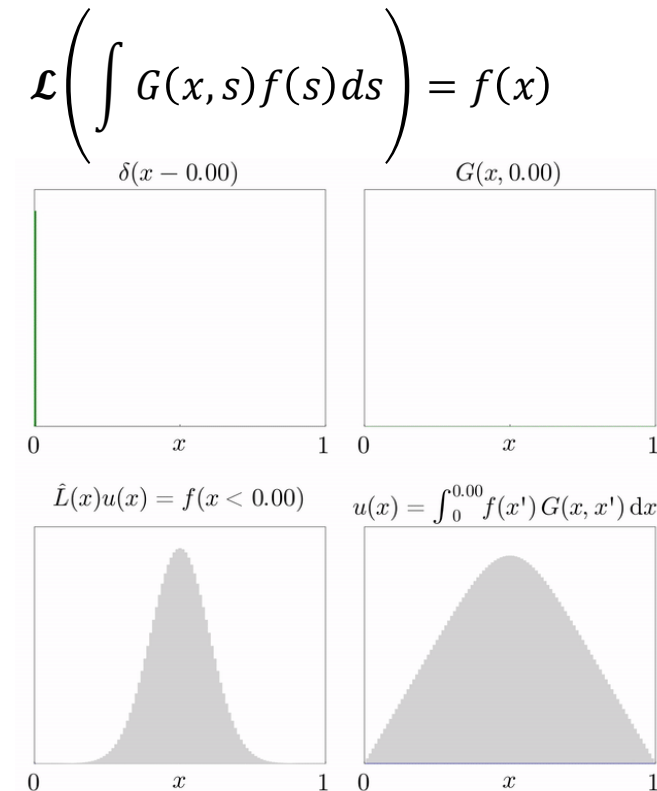
is **A solution** to  
(A *unique* solution is achieved via BC)

$$\mathcal{L}u(x) = f(x)$$

- **Goal?** find the Green's function  $G$  that satisfies:

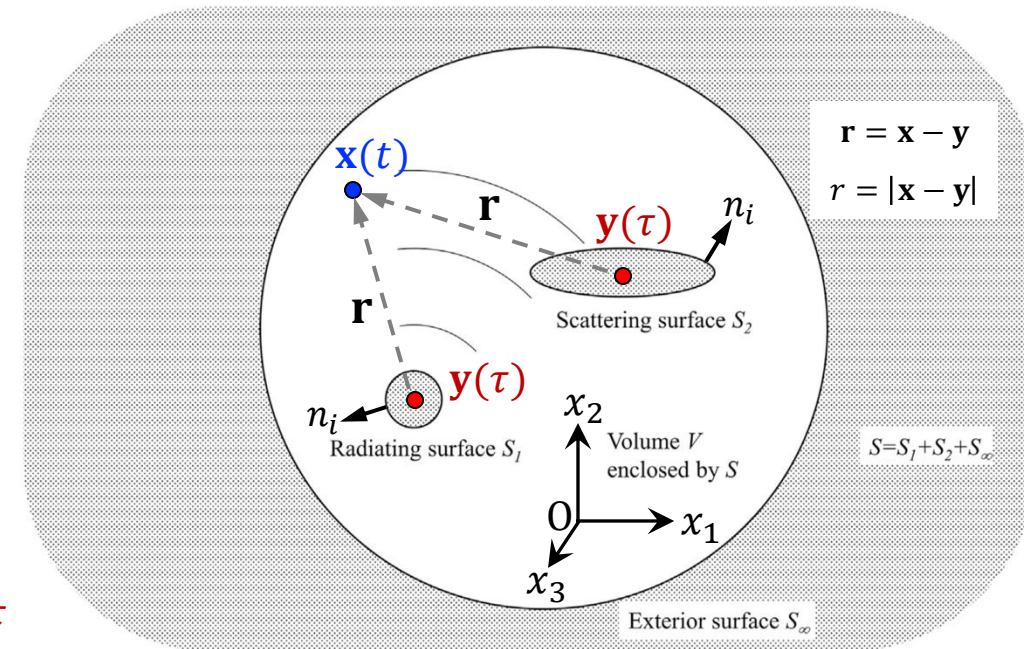
$$\mathcal{L}G(x, s) = \delta(x - s)$$

- $G(x, s)$  must be *continuous* in  $x$  and  $s$
- For  $x \neq s$ ,  $\mathcal{L}G(x, s) = 0$
- **Symmetry** or "**Reciprocity Principle**",  $G(x, s) = G(s, x)$
- Green's function is often called the *fundamental solution* associated with  $\mathcal{L}$
- Not every operator  $\mathcal{L}$  admits a Green's function...
- The integral for  $u(x)$  may be quite difficult to evaluate...



# A general solution in the presence of surfaces

- ❑ In most aerodynamics and aeroacoustics applications, **surfaces are involved**, which can:
  - Scatter sound, or
  - Be a **source of sound**
- ❑ To find a general solution for a CV with surfaces inside, we invoke the **method of Green's functions**
- ❑ Consider a **CV with a uniform stationary medium** that **satisfies the linear wave equation**
  - CV can include surfaces that can **generate** and reflect sound
  - Surface sources are defined by space coordinate **y** and time **τ**
- ❑ **We wish to determine the acoustic pressure  $p'(\mathbf{x}, t)$  at observer location  $\mathbf{x}$  at time  $t$  from waves in the fluid and the surface sources**
- ❑ We first consider the **time-domain homogeneous wave eq.** for the pressure in terms of **y** and **τ**:



$$\square p'(\mathbf{y}, \tau) = 0$$

$$\frac{1}{c_0^2} \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial \tau^2} - \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial y_i^2} = 0$$

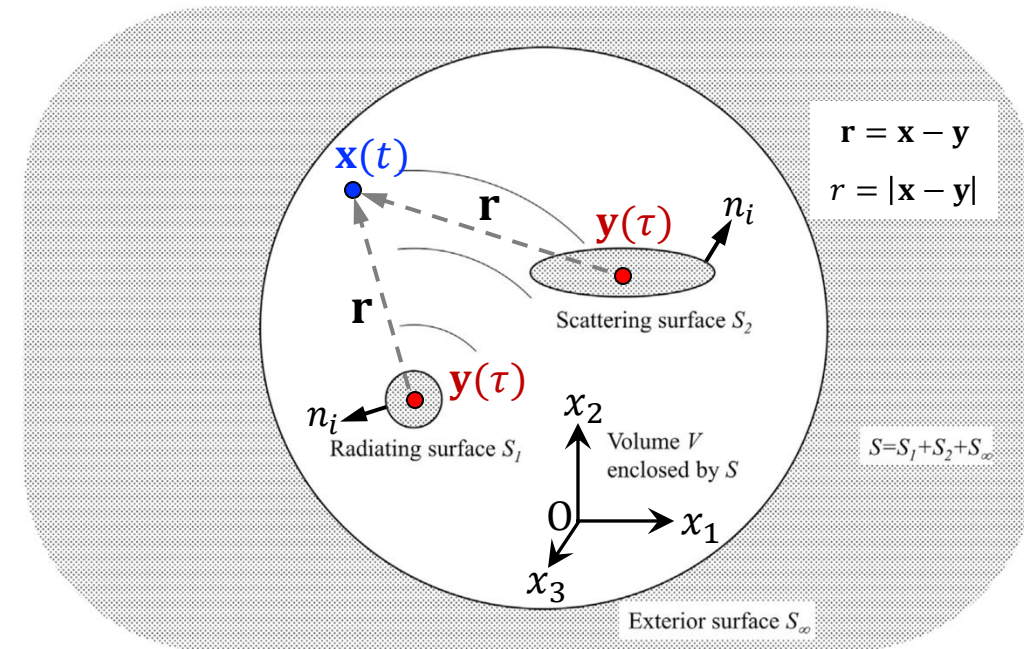
# A general solution in the presence of surfaces

- The **Green's function**  $G$  is introduced – defined as  
A solution to the **inhomogeneous wave equation**:

$$\square G(\mathbf{x}, t | \mathbf{y}, \tau) = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau)$$

$$\frac{1}{c_0^2} \frac{\partial^2 G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial \tau^2} - \frac{\partial^2 G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau)$$

- $G$  is the acoustic field (solution of the wave eq.) generated at position  $\mathbf{x}$  of an **observer** at time  $t$  by a **pulse**  $\delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau)$  emitted by a **source** at  $\mathbf{y}$  at time  $\tau$ 
  - $G$  is a function of  $\mathbf{y}$  and time  $\tau$  but also depends on the observer position  $\mathbf{x}$  and time  $t \rightarrow$  solution has a functional form  $G(\mathbf{x}, t | \mathbf{y}, \tau)$
  - Due to linearity, once we know the response to a delta source, we can build the response to everything else by **convolution**



$G$  is the solution of the **same linear operator**  $\square$  acting on  $p'$ , but with a point-impulse source

Same BC we satisfied for  $p'$

$$\frac{1}{c_0^2} \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial \tau^2} - \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial y_i^2} = 0$$

# Dirac delta function

- A generalized function that is very helpful for solving dynamic problems

➤ It is sometimes defined as the *unit impulse function*

$$\delta(x) = \begin{cases} \infty, & \text{if } x = 0, \\ 0, & \text{if } x \neq 0. \end{cases} \quad \text{such that: } \int_{-\infty}^{\infty} \delta(x) dx = 1 \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$$

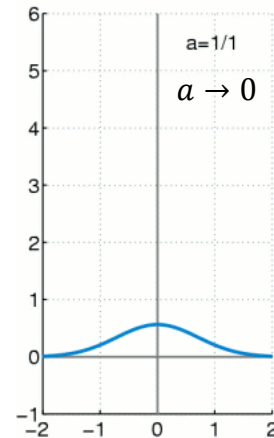
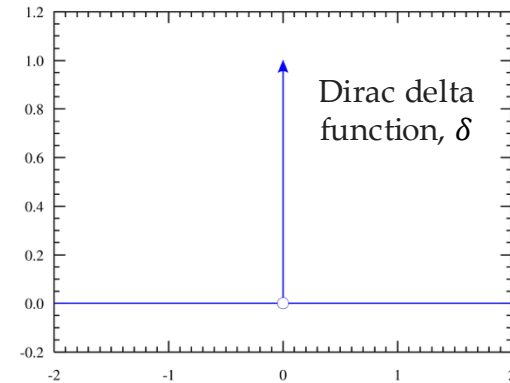
- In aeroacoustics, we often use this **time-delayed Dirac delta function**:

$$\delta(t - \tau) = \begin{cases} \infty, & \text{if } t = \tau, \\ 0, & \text{if } t \neq \tau. \end{cases} \quad \text{such that: } \int_{-\infty}^{\infty} \delta(t - \tau) d\tau = 1, \quad -\infty < t < \infty$$

- **Sifting property** – for any function  $G(t)$  continuous at  $\tau$ :

$$\int_{-\infty}^{\infty} G(t) \delta(t - \tau) dt = G(\tau)$$

➤ The sifting property of the Dirac delta function allows to measure the value of  $G(t)$  at  $t = \tau$



$$\delta_a(x) = \frac{1}{|a|\sqrt{\pi}} e^{-(x/a)^2}$$

Approx. as a sequence of zero-centered Gaussian distributions with variance tending to zero ( $a \rightarrow 0$ )



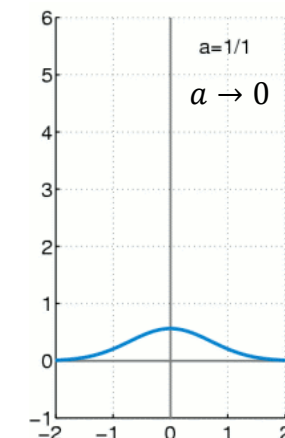
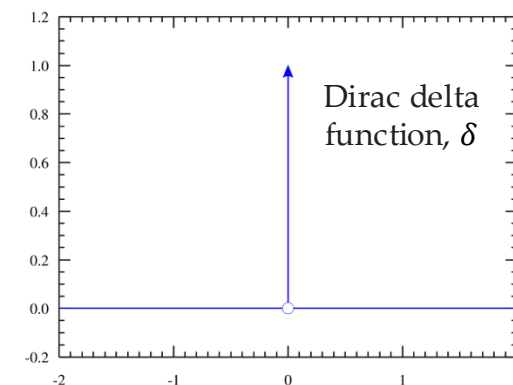
# Dirac delta function

## □ Composition with a function

- $g(x)$  – smooth function (continuously differentiable)
- $g(x)$  has  $n$  number of zeros  $x_i = [x_1, \dots, x_n]$ , such that  $g(x_i) = 0$  and  $g'(x_i) \neq 0$

$$\int_{-\infty}^{\infty} f(x) \delta(g(x)) dx = \sum_{i=1}^n \frac{f(x_i)}{|g'(x_i)|}$$

$$g'(x_i) = \left. \frac{dg}{dx} \right|_{x_i}$$



$$\delta_a(x) = \frac{1}{|a|\sqrt{\pi}} e^{-(x/a)^2}$$

Approx. as a sequence of zero-centered Gaussian distributions with variance tending to zero ( $a \rightarrow 0$ )

# Dirac delta function

- When the argument of the Dirac is a **vector**, we define it so that:

$$\delta(\mathbf{x} - \mathbf{y}) = \begin{cases} \infty, & \text{if } \mathbf{x} = \mathbf{y}, \\ 0, & \text{if } \mathbf{x} \neq \mathbf{y}. \end{cases} \quad \text{such that: } \int_V \delta(\mathbf{x} - \mathbf{y}) dV(\mathbf{y}) = \begin{cases} 1, & \text{if } \mathbf{x} \in V, \\ 0, & \text{if } \mathbf{x} \notin V. \end{cases}$$

- Sifting property** – for any function  $G(\mathbf{x})$  continuous at  $\mathbf{y}$ :

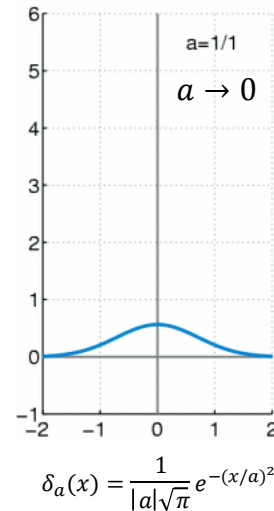
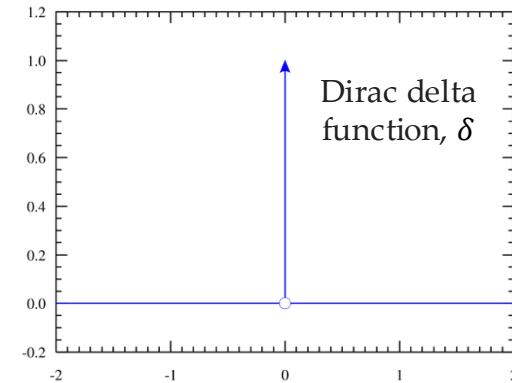
$$\int_{\mathbb{R}^3} G(\mathbf{x}) \delta(\mathbf{x} - \mathbf{y}) d^3\mathbf{x} = G(\mathbf{y})$$

- Isotropic Gaussian approximation:**  $\delta_\epsilon(\mathbf{x} - \mathbf{y}) \approx \frac{1}{(\epsilon\sqrt{\pi})^3} e^{-(|\mathbf{x}-\mathbf{y}|/\epsilon)^2}$

- For **Cartesian coordinates**, we have:  $\delta(\mathbf{x} - \mathbf{y}) = \delta(x_1 - y_1)\delta(x_2 - y_2)\delta(x_3 - y_3)$

- For **spherical coordinates**, we have:  $\delta(\mathbf{x} - \mathbf{y}) = \lim_{\epsilon^+ \rightarrow 0} \left( \frac{e^{-(r/\epsilon)^2}}{2\pi^{3/2}\epsilon r^2} \right) \quad r = |\mathbf{x} - \mathbf{y}|$

$$\delta(\mathbf{x} - \mathbf{y}) = \frac{\delta(r)}{4\pi r^2}$$



Approx. as a sequence of zero-centered Gaussian distributions with variance tending to zero ( $a \rightarrow 0$ )

# Solution process

□ We look for a solution to the linear wave equation...

□ **Our approach:**

$$\left. \begin{array}{l} \text{➤ Take Green's function equation:} \quad \frac{1}{c_0^2} \frac{\partial^2 G}{\partial \tau^2} - \frac{\partial^2 G}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau) \quad \times p'(\mathbf{y}, \tau) \\ \text{➤ Take the linear wave equation:} \quad \frac{1}{c_0^2} \frac{\partial^2 p'}{\partial \tau^2} - \frac{\partial^2 p'}{\partial y_i^2} = 0 \quad \times G(\mathbf{x}, t | \mathbf{y}, \tau) \end{array} \right\} (-) \text{ Subtract}$$

$$\Rightarrow \frac{1}{c_0^2} \left( p' \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 p'}{\partial \tau^2} \right) - \left( p' \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 p'}{\partial y_i^2} \right) = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau) p'(\mathbf{y}, \tau)$$

□ Integrate over volume  $V$  and time  $\tau$ :

$$\int_V \int_{-T}^T \left[ \frac{1}{c_0^2} \left( p' \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 p'}{\partial \tau^2} \right) - \left( p' \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 p'}{\partial y_i^2} \right) \right] d\tau dV(\mathbf{y}) = p'(\mathbf{x}, t)$$

$$\int_{-T}^T p'(\tau) \delta(t - \tau) dt = p'(t)$$

$$\int_V p'(\mathbf{y}) \delta(\mathbf{x} - \mathbf{y}) dV = p'(\mathbf{x})$$

➤ Providing  $\mathbf{x}$  is inside  $V$  and  $-T < t < T$



# Solution process

$$\int_V \int_{-T}^T \left[ \underbrace{\frac{1}{c_0^2} \left( p' \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 p'}{\partial \tau^2} \right)}_{\text{I}} - \underbrace{\left( p' \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 p'}{\partial y_i^2} \right)}_{\text{II}} \right] d\tau dV(\mathbf{y}) = p'(\mathbf{x}, t)$$

□ **Term I:**  $p' \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 p'}{\partial \tau^2} = \frac{\partial}{\partial \tau} \left( p' \frac{\partial G}{\partial \tau} - G \frac{\partial p'}{\partial \tau} \right) \Rightarrow \int_V \int_{-T}^T \frac{\partial}{\partial \tau} \left( p' \frac{\partial G}{\partial \tau} - G \frac{\partial p'}{\partial \tau} \right) d\tau dV(\mathbf{y}) = \int_V \left[ p' \frac{\partial G}{\partial \tau} - G \frac{\partial p'}{\partial \tau} \right]_{\tau=-T}^{\tau=T} dV(\mathbf{y})$

- Choose these **initial conditions**:  $\begin{cases} p' = 0, & \text{if } \tau = -T, \\ \partial p' / \partial \tau = 0, & \text{if } \tau = -T. \end{cases} \Rightarrow \left[ p' \frac{\partial G}{\partial \tau} - G \frac{\partial p'}{\partial \tau} \right]_{\tau=-T} = 0$
- Enforce **Causality condition**:  $\begin{cases} G(\mathbf{x}, t | \mathbf{y}, \tau) = 0, & \text{if } \tau \geq t, \\ \partial G(\mathbf{x}, t | \mathbf{y}, \tau) / \partial \tau = 0, & \text{if } \tau \geq t. \end{cases} \xrightarrow{\text{if } t < T} \left[ p' \frac{\partial G}{\partial \tau} - G \frac{\partial p'}{\partial \tau} \right]_{\tau=T} = 0$

Sound heard at time  $t$  must be produced at time  $\tau < t$

Thus:

$$\text{I} = \int_V \int_{-T}^T \frac{1}{c_0^2} \left( p' \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 p'}{\partial \tau^2} \right) d\tau dV(\mathbf{y}) = 0$$

# Solution process

$$\int_V \int_{-T}^T \left[ \underbrace{\frac{1}{c_0^2} \left( p' \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 p'}{\partial \tau^2} \right)}_{\text{I}} - \underbrace{\left( p' \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 p'}{\partial y_i^2} \right)}_{\text{II}} \right] d\tau dV(\mathbf{y}) = p'(\mathbf{x}, t)$$

□ **Term II:**  $p' \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 p'}{\partial y_i^2} = \frac{\partial}{\partial y_i} \left( p' \frac{\partial G}{\partial y_i} - G \frac{\partial p'}{\partial y_i} \right) \Rightarrow \int_V \left( p' \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 p'}{\partial y_i^2} \right) dV(\mathbf{y}) = \int_V \frac{\partial}{\partial y_i} \left( p' \frac{\partial G}{\partial y_i} - G \frac{\partial p'}{\partial y_i} \right) dV(\mathbf{y})$

➤ Use **divergence theorem**:

$$\int_V \frac{\partial}{\partial y_i} \left( p' \frac{\partial G}{\partial y_i} - G \frac{\partial p'}{\partial y_i} \right) dV(\mathbf{y})$$

Green's identity  $\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \int_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} dS$

$$\int_V \left( p' \frac{\partial G}{\partial y_i} - G \frac{\partial p'}{\partial y_i} \right) n_i dS(\mathbf{y})$$

➤  $n_i$  – unit vector normal to the surface of  $\mathbf{y}$  pointing **into** the volume

Thus:

$$\boxed{\text{II} = - \int_S \left( p' \frac{\partial G}{\partial y_i} - G \frac{\partial p'}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y})}$$

# A general solution in the presence of surfaces

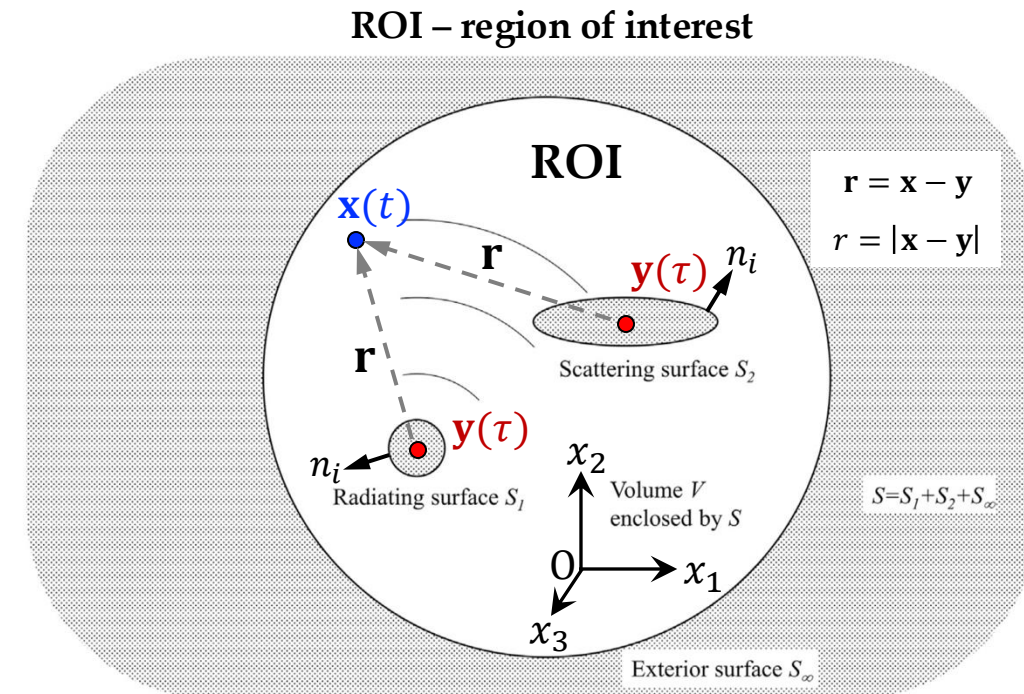
$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$$

## □ NOTE!

- Surface location  $\mathbf{y}$  is taken **fixed in time**
- Surface  $\mathbf{y}$  can be **porous** to *allow* flow through the surface, so **pressure gradients** normal to the surface are **possible**

## □ Sommerfeld radiation condition

- There can only be outgoing waves on the exterior boundary  $S_\infty \rightarrow$  **sound waves only radiate out**
- **No sound reflects back to the observer from  $S_\infty$**   
 $\rightarrow$  exterior surface excluded from surface integral



# A general solution in the presence of surfaces

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$$

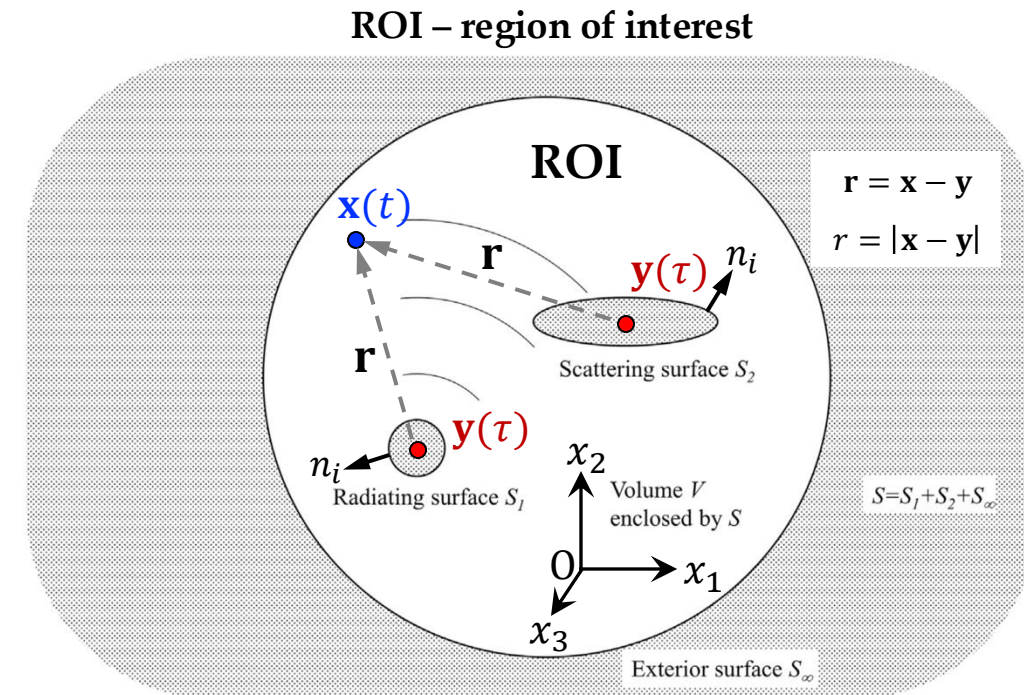
## □ Key take aways:

Any sound field satisfying  $\square p' = 0$  can be evaluated from knowledge of  $p'$  and  $\partial p' / \partial y_i$  on surfaces  $S$  bounding the ROI

A Green's function is to be found, which must satisfy  $\square G = \delta(\mathbf{x} - \mathbf{y})\delta(t - \tau)$  and a causality condition ( $t > \tau$ )

The result can be used for both:

1. **Sound radiation** – fluid motion at the surface and/or surface pressure is *prescribed*
2. **Sound scattering** – passive BCs in terms  $p'$  and  $\partial p' / \partial y_i$  are *defined* on the scattering surfaces





# A general solution in the presence of surfaces

## Frequency domain

- For many applications, interest is in the frequency content of a non-sinusoidal signal

➤ Can be obtained from the Fourier transform

$$\tilde{p}(\mathbf{x}, \omega) = \frac{1}{2\pi} \int_{-T}^{+T} p'(\mathbf{x}, t) e^{i\omega t} dt$$

$$p'(\mathbf{x}, t) = \int_{-\infty}^{+\infty} \tilde{p}(\mathbf{x}, \omega) e^{-i\omega t} d\omega$$

inverse

- Fourier transform of the *homogeneous wave eq.* with respect to the variable  $\tau$  yields the **Helmholtz eq.**

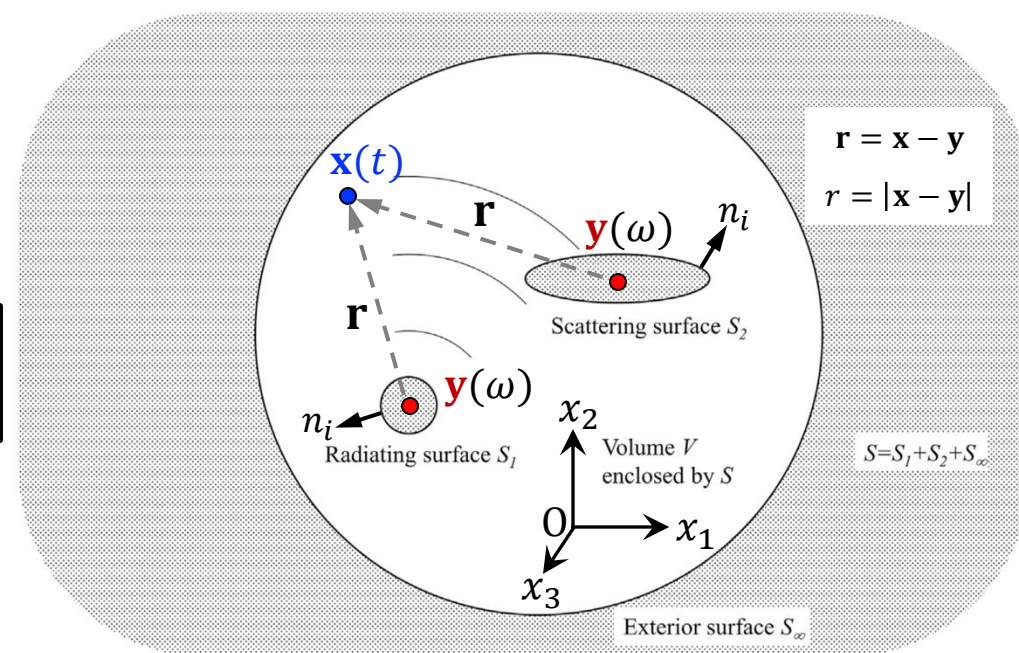
$$\Rightarrow \frac{\partial^2 \tilde{p}(\mathbf{y}, \omega)}{\partial y_i^2} + k^2 \tilde{p}(\mathbf{y}, \omega) = 0$$

Acoustic wavenumber  
 $k = \omega/c_0$

➤ Frequency-domain homogeneous wave equation for the pressure in terms of  $\mathbf{y}$  and  $\omega$

- We wish to determine the acoustic pressure  $\tilde{p}(\mathbf{x}, \omega)$  at observer location  $\mathbf{x}$  for acoustic waves of frequency  $\omega$

Green's function



Time-domain

$$\square p'(\mathbf{y}, \tau) = 0$$

$$\frac{1}{c_0^2} \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial \tau^2} - \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial y_i^2} = 0$$

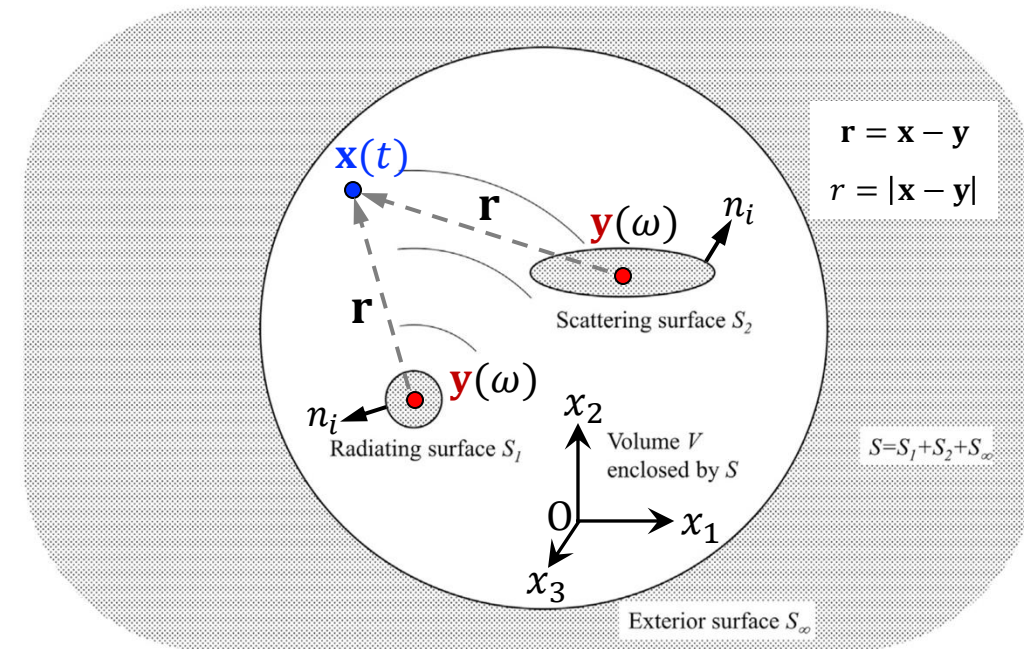
# A general solution in the presence of surfaces

## Frequency domain

- The Green's function  $G$  is introduced – defined as A solution to the *inhomogeneous equation*:

$$\frac{\partial^2 \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i^2} + k^2 \tilde{G}(\mathbf{x}|\mathbf{y}) = -\delta(\mathbf{x} - \mathbf{y})$$

- $G$  is the acoustic field generated at position  $\mathbf{x}$  of an **observer** at some angular frequency  $\omega$  by a **pulse**  $\delta(\mathbf{x} - \mathbf{y})$  emitted by a **source** at  $\mathbf{y}$ 
  - $G$  is a function of  $\mathbf{y}$  and observer position  $\mathbf{x} \rightarrow$  solution has a functional form  $G(\mathbf{x}|\mathbf{y})$
  - As before, once we know the response to a delta source, we can build the response to everything else by *convolution*



$G$  is the solution of the **same linear operator** acting on  $\tilde{p}$ , but with a point-impulse source

Same BC we satisfied for  $\tilde{p}$

$$\frac{\partial^2 \tilde{p}(\mathbf{y}, \omega)}{\partial y_i^2} + k^2 \tilde{p}(\mathbf{y}, \omega) = 0$$

# Solution process

## Frequency domain

### □ Following a similar procedure as before...

$$\begin{array}{ll}
 \text{➤ Take Green's function equation:} & \frac{\partial^2 \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i^2} + k^2 \tilde{G}(\mathbf{x}|\mathbf{y}) = -\delta(\mathbf{x} - \mathbf{y}) \quad \times \tilde{p}(\mathbf{y}, \omega) \\
 \text{➤ Take the linear wave equation:} & \frac{\partial^2 \tilde{p}(\mathbf{y}, \omega)}{\partial y_i^2} + k^2 \tilde{p}(\mathbf{y}, \omega) = 0 \quad \times \tilde{G}(\mathbf{x}|\mathbf{y})
 \end{array}
 \left. \vphantom{\begin{array}{l} \text{➤ Take Green's function equation:} \\ \text{➤ Take the linear wave equation:} \end{array}} \right\} (-) \text{ Subtract}$$

$$\Rightarrow \tilde{p} \frac{\partial^2 \tilde{G}}{\partial y_i^2} - \tilde{G} \frac{\partial^2 \tilde{p}}{\partial y_i^2} = -\delta(\mathbf{x} - \mathbf{y}) \tilde{p}(\mathbf{y}, \omega)$$

### □ Integrate over volume $V$ :

$$\tilde{p}(\mathbf{x}, \omega) = - \int_V \left( \tilde{p}(\mathbf{y}, \omega) \frac{\partial^2 \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i^2} - \tilde{G}(\mathbf{x}|\mathbf{y}) \frac{\partial^2 \tilde{p}(\mathbf{y}, \omega)}{\partial y_i^2} \right) dV(\mathbf{y})$$

➤ Providing  $\mathbf{x}$  is inside  $V$

$$\int_V p'(\mathbf{y}) \delta(\mathbf{x} - \mathbf{y}) dV = p'(\mathbf{x})$$



# Solution process

## Frequency domain

$$\tilde{p}(\mathbf{x}, \omega) = - \int_V \left( \tilde{p}(\mathbf{y}, \omega) \frac{\partial^2 \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i^2} - \tilde{G}(\mathbf{x}|\mathbf{y}) \frac{\partial^2 \tilde{p}(\mathbf{y}, \omega)}{\partial y_i^2} \right) dV(\mathbf{y})$$

□ Note:  $\tilde{p} \frac{\partial^2 \tilde{G}}{\partial y_i^2} - \tilde{G} \frac{\partial^2 \tilde{p}}{\partial y_i^2} = \frac{\partial}{\partial y_i} \left( \tilde{p} \frac{\partial \tilde{G}}{\partial y_i} - \tilde{G} \frac{\partial \tilde{p}}{\partial y_i} \right) \Rightarrow \int_V \left( \tilde{p} \frac{\partial^2 \tilde{G}}{\partial y_i^2} - \tilde{G} \frac{\partial^2 \tilde{p}}{\partial y_i^2} \right) dV(\mathbf{y}) = \int_V \frac{\partial}{\partial y_i} \left( \tilde{p} \frac{\partial \tilde{G}}{\partial y_i} - \tilde{G} \frac{\partial \tilde{p}}{\partial y_i} \right) dV(\mathbf{y})$

➤ Use **divergence theorem**:  $\int_V \frac{\partial}{\partial y_i} \left( \tilde{p} \frac{\partial \tilde{G}}{\partial y_i} - \tilde{G} \frac{\partial \tilde{p}}{\partial y_i} \right) dV(\mathbf{y}) = - \int_S \left( \tilde{p} \frac{\partial \tilde{G}}{\partial y_i} - \tilde{G} \frac{\partial \tilde{p}}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y})$

Green's identity  $\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = - \int_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} dS$

➤  $n_i$  – unit vector normal to the surface of  $\mathbf{y}$  pointing **into** the volume



$$\tilde{p}(\mathbf{x}, \omega) = \int_S \left( \tilde{p}(\mathbf{y}, \omega) \frac{\partial \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i} - \tilde{G}(\mathbf{x}|\mathbf{y}) \frac{\partial \tilde{p}(\mathbf{y}, \omega)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y})$$

Equivalence to  $p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t|\mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t|\mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$

# A general solution in the presence of surfaces

## Frequency domain

$$\tilde{p}(\mathbf{x}, \omega) = \int_S \left( \tilde{p}(\mathbf{y}, \omega) \frac{\partial \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i} - \tilde{G}(\mathbf{x}|\mathbf{y}) \frac{\partial \tilde{p}(\mathbf{y}, \omega)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y})$$

□ Taking the inverse Fourier transform  $p'(\mathbf{x}, t) = \int_{-\infty}^{+\infty} \tilde{p}(\mathbf{x}, \omega) e^{-i\omega t} d\omega$  and substituting  $\tilde{p}(\mathbf{y}, \omega) = \frac{1}{2\pi} \int_{-T}^{+T} p'(\mathbf{y}, \tau) e^{i\omega \tau} d\tau$

$$\Rightarrow p'(\mathbf{x}, t) = \int_{-\infty}^{+\infty} \int_S \left\{ \frac{1}{2\pi} \int_{-T}^{+T} \left( p'(\mathbf{y}, \tau) \left[ \frac{\partial \tilde{G}(\mathbf{x}|\mathbf{y})}{\partial y_i} \right]_{\omega} - [\tilde{G}(\mathbf{x}|\mathbf{y})]_{\omega} \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) e^{-i\omega(t-\tau)} d\tau \right\} dS(\mathbf{y}) d\omega$$

➤ Comparing this result with the previous time-domain solution for  $p'(\mathbf{x}, t)$  reveals the following **relationship between Green's function in the time- and frequency-domains**

Time-domain solution:  $p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t|\mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t|\mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$



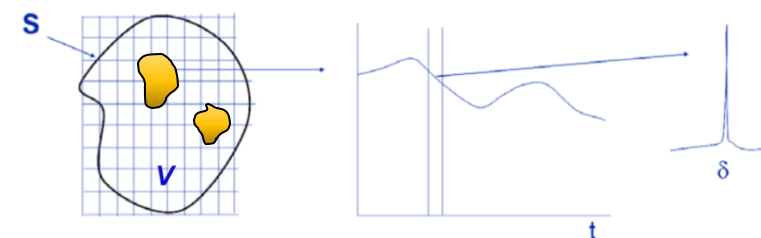
$$G(\mathbf{x}, t|\mathbf{y}, \tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} [\tilde{G}(\mathbf{x}|\mathbf{y})]_{\omega} e^{-i\omega(t-\tau)} d\omega$$

$$\tilde{G}(\mathbf{x}|\mathbf{y}) = \int_{-T}^{+T} G(\mathbf{x}, t|\mathbf{y}, \tau) e^{i\omega t} dt \quad \text{inverse}$$

# More general solution in the presence of surfaces

- A **more general solution** can be achieved with **arbitrary initial conditions** ( $\tau = t_0$ , causality is still enforced) and a **source term** in the wave eq. (*inhomogeneous* form) :

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = Q(\mathbf{x}, t)$$



$$p'(\mathbf{x}, t) = \int_{t_0}^t \int_V Q(\mathbf{y}, \tau) G(\mathbf{x}, t | \mathbf{y}, \tau) dV d\tau + \int_{t_0}^t \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau - \frac{1}{c_0^2} \int_V \left[ p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial \tau} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=t_0} dV$$

Convolution of the pulse and  $G$

Difference between boundary conditions on the surface and  $G$

Initial conditions at  $t_0$

# Free-field Green's function

- Given a general solution to the homogeneous wave eq.:  $p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$
- We can now analyze **any radiating surface**
- If there is an imposed flow  $v_i$  at the surface, then the pressure gradient term can be obtained from the acoustic momentum eq.:

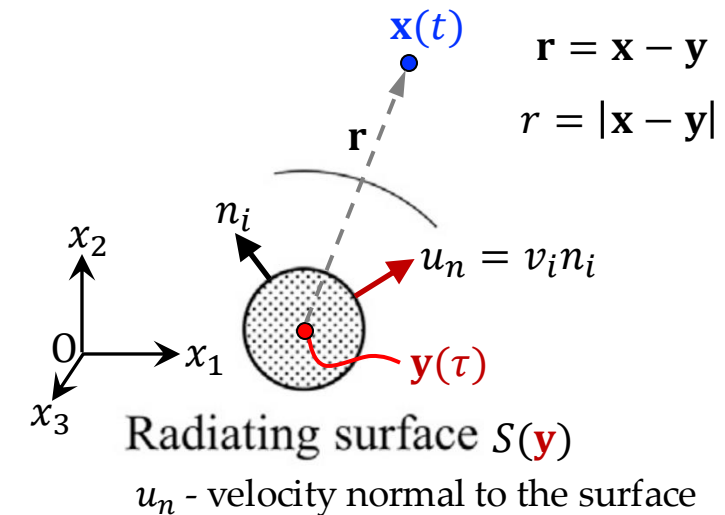
$$\text{Acoustic momentum eq.} \quad \rho_0 \frac{\partial v_i}{\partial t} + \frac{\partial p'}{\partial x_i} = 0 \quad \Rightarrow \quad \frac{\partial p'}{\partial y_i} n_i = \mathbf{n} \cdot \nabla p' = -\rho_0 \frac{\partial v_i}{\partial \tau} n_i$$

- If the fixed surface of integration is used to represent a **solid surface that is vibrating**, then the imposed flow can be taken to be the vibration velocity on the surface
- Therefore, we can obtain:

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left( \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau$$

Sound radiated from **force per unit area**  $p'n_i$  exerted on fluid by surface

Sound radiated by **imposed velocity**

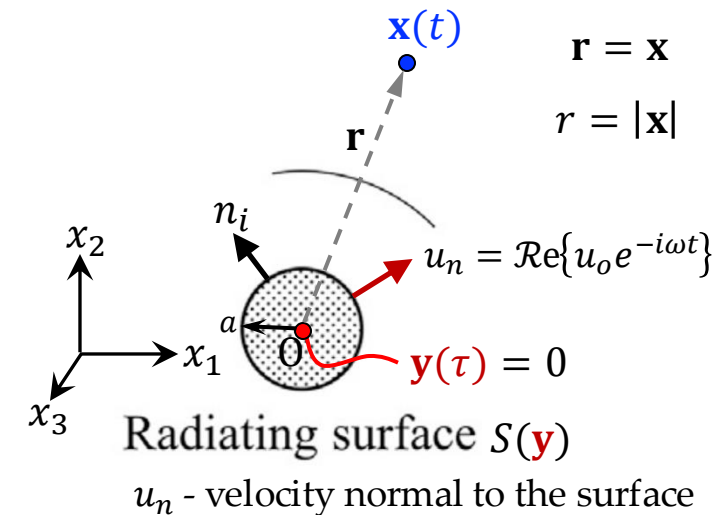


# Free-field Green's function

- ❑ To evaluate the integrals, **we need to specify the Green's function**
- ❑ We already know the solution for a small pulsating sphere...
- ❑ Consider a small spherical surface with its center at  $\mathbf{y} = 0$ 
  - Normal velocity on the surface is given by  $u_n = \mathcal{R}\{u_o e^{-i\omega t}\}$
- ❑ If we allow the radius of the sphere source to become smaller and smaller ( $a \rightarrow 0$ ), then eventually the spatial dependence of  $G$  and  $\partial G / \partial \mathbf{y}_i$  will become **not important**
  - In a sense, we assume a **compact sphere source** ( $ka \ll 1$ )
  - If observer is not on the sphere (i.e.,  $\mathbf{x} \neq \mathbf{y}$ ), and  $G(\mathbf{x}|\mathbf{y})$  is a smooth function of  $\mathbf{y}$ , we can expand it via Taylor series about the center  $\mathbf{y} = 0$ :

$$G(\mathbf{x}|\mathbf{y}) = G(\mathbf{x}|0) + \mathbf{y}_i \left. \frac{\partial G}{\partial \mathbf{y}_i} \right|_{\mathbf{y}=0} + \mathcal{O}(|\mathbf{y}|^2) \quad \frac{\partial G(\mathbf{x}|\mathbf{y})}{\partial \mathbf{y}_i} = \left. \frac{\partial G}{\partial \mathbf{y}_i} \right|_{\mathbf{y}=0} + \mathbf{y}_j \left. \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right|_{\mathbf{y}=0} + \mathcal{O}(|\mathbf{y}|^2)$$

- On the sphere  $|\mathbf{y}| = a$ , and so we obtain:  $\Delta G \sim a \left. \frac{\partial G}{\partial \mathbf{y}_i} \right|_{\mathbf{y}=0}$  and  $\Delta \left( \frac{\partial G}{\partial \mathbf{y}_i} \right) \sim a \left. \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right|_{\mathbf{y}=0}$
- But, as we take  $a \rightarrow 0$ , we obtain:  $G(\mathbf{x}|\mathbf{y}) = \mathbb{C}$  and  $\partial G / \partial \mathbf{y}_i = \mathbb{C}$



# Free-field Green's function

- Therefore, assuming a vanishingly small sphere, we can take  $G$  and  $\partial G / \partial y_i$  out of the surface integral (assuming they are constants), yielding:

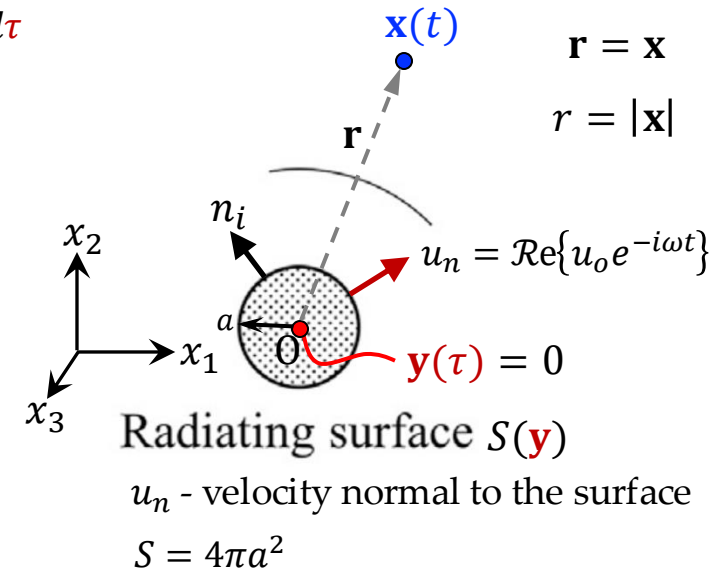
$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left( p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left( \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau$$

$$\Rightarrow p'(\mathbf{x}, t) = \int_{-T}^T \left[ \frac{\partial G}{\partial y_i} \right]_{y_i=0} \underbrace{\int_S p'(\mathbf{y}, \tau) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau}_{\text{Net force exerted on the fluid by the sphere is zero since pressure is constant on the surface}} + \text{Re} \left\{ \int_{-T}^T [G]_{y_i=0} \int_S (-i\omega \rho_0 u_o e^{-i\omega\tau}) dS(\mathbf{y}) d\tau \right\}$$

Net force exerted on the fluid by the sphere is **zero** since pressure is **constant** on the surface

$$\left[ \frac{\partial p'}{\partial r} \right]_{r=a} = -\rho_0 \frac{\partial v_i}{\partial \tau} n_i = \text{Re} \{ i\omega \rho_0 u_o e^{-i\omega\tau} \} = \mathbb{C} \Rightarrow \int_S (-i\omega \rho_0 u_o e^{-i\omega\tau}) dS(\mathbf{y}) = -i\omega \rho_0 u_o S e^{-i\omega\tau}$$

$$\Rightarrow p'(\mathbf{x}, t) = \text{Re} \left\{ \int_{-T}^T (-i\omega \rho_0 u_o S e^{-i\omega\tau}) [G(\mathbf{x}, t | \mathbf{y}, \tau)]_{y_i=0} d\tau \right\}$$





# Free-field Green's function

- Let us **compare** the result we obtain to the **monopole sound field** we obtained previously

$$p'(\mathbf{x}, t) = \text{Re} \left\{ \int_{-T}^T (-i\omega\rho_0 u_0 S e^{-i\omega\tau}) [G(\mathbf{x}, t | \mathbf{y}, \tau)]_{\mathbf{y}_i=0} d\tau \right\}$$

$$p'(r) = -\frac{i\omega\rho_0 u_0 S e^{-i(\omega\tau - kr)}}{4\pi r}$$

- We reveal **Free-Field Green's function**  $G_0$ , given  $r = |\mathbf{x} - \mathbf{y}| > 0$ :

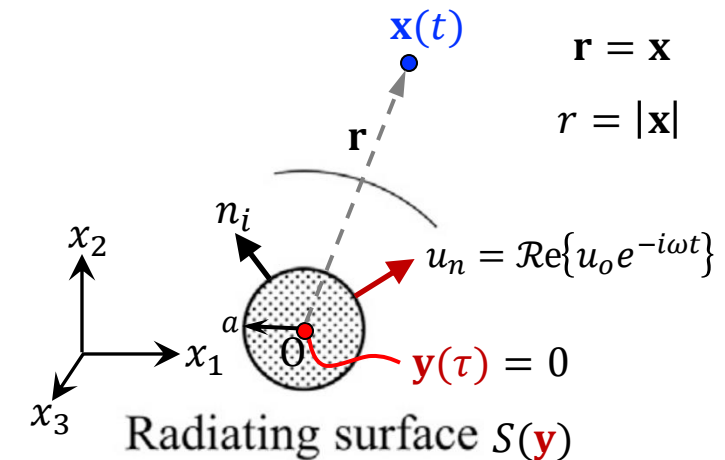
**Source** is located at  $\mathbf{y} = 0$

$$G_0(r, t - \tau) = \frac{\delta(t - r/c_0 - \tau)}{4\pi r}$$

**Source** is located at  $\mathbf{y}$

$$G_0(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_0 - \tau)}{4\pi |\mathbf{x} - \mathbf{y}|}$$

- Acoustic field at the observer location  $\mathbf{x}$  caused by an infinitely small volume displacement source (*monopole*) in an **infinite domain**



$u_n$  - velocity normal to the surface  
 $S = 4\pi a^2$

Monopole sound field

$$\hat{p}(r) = -\frac{i\omega\rho_0 u_0 S e^{ikr}}{4\pi r} \quad k = \omega/c_0$$

$$p'(r) = \hat{p}(r) e^{-i\omega t}$$

# Free-field Green's function

- The free-field Green's function  $G_o$  is the acoustic field (solution of a wave eq.) generated at position  $\mathbf{x}$  of an observer at time  $t$  by a pulse  $\delta(\mathbf{x} - \mathbf{y})\delta(t - \tau)$  emitted at  $\mathbf{y}$  at time  $\tau$ , provided that  $r = |\mathbf{x} - \mathbf{y}| > 0$

$$\begin{aligned} \mathbf{r} &= \mathbf{x} - \mathbf{y} \\ r &= |\mathbf{x} - \mathbf{y}| \end{aligned} \quad G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_o - \tau)}{4\pi|\mathbf{x} - \mathbf{y}|} \quad \text{is A solution to} \quad \frac{1}{c_o^2} \frac{\partial^2 G_o}{\partial \tau^2} - \frac{\partial^2 G_o}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y})\delta(t - \tau)$$

- The Green's free-field function  $G_o$  satisfies the **principle of causality** when  $r > 0$

$$\text{For } t < \tau: \quad G_o = \frac{\partial G_o}{\partial \tau} = 0 \quad \text{because } r/c_o > 0$$

- Since it depends only on  $r$ , it also satisfies the **symmetry property**:

$$\frac{\partial G_o}{\partial x_i} = -\frac{\partial G_o}{\partial y_i} \quad \text{because } \frac{\partial r}{\partial x_i} = \frac{x_i - y_i}{r} = -\frac{\partial r}{\partial y_i}$$

Bringing the **observer** closer to the **source** has the same effect as bringing the **observer** closer to the **source**!

# Free-field Green's function

- Let's define the gradient of  $G_o$

$$r = |\mathbf{x} - \mathbf{y}| \quad \frac{\partial G_o}{\partial y_i} = \left( \frac{x_i - y_i}{r} \right) \left( \frac{\dot{\delta}(t - r/c_o - \tau)}{4\pi r c_o} + \frac{\delta(t - r/c_o - \tau)}{4\pi r^2} \right) \quad \dot{\delta} = \frac{\partial \delta}{\partial t}$$

- Note that the field from the term above has a **cosine directionality** built in and,
- Also, the field consists of both a **near-field** term and a **far-field** term

- By using the Green's function, we can evaluate the time integral in the pressure equation  $p'(\mathbf{x}, t)$ 
  - We integrate the time derivative of the Dirac delta function by parts

$$p'(\mathbf{x}, t) = \int_S \left[ \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} n_i + \frac{p'(\mathbf{y}, \tau) n_i c_o}{|\mathbf{x} - \mathbf{y}|} \right]_{\tau=\tau^*} \frac{(x_i - y_i) dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|} + \int_S \left[ \rho_o \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=\tau^*} \frac{dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|}$$

Sound generated by surface **loading**

Sound generated by **imposed flow** at the surface

Retarded time

$$\tau = t - \frac{|\mathbf{x} - \mathbf{y}|}{c_o}$$

**source time** or **emission time**

# Free-field Green's function

□ In the acoustic far-field,  $|\mathbf{x}| \gg |\mathbf{y}|$

$$p'(\mathbf{x}, t) \approx \underbrace{\frac{x_i}{4\pi|\mathbf{x}|^2 c_0} \int_S \left[ \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} n_i \right]_{\tau=\tau^*} dS(\mathbf{y})}_{\text{Dipole}} + \underbrace{\frac{1}{4\pi|\mathbf{x}|} \int_S \left[ \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=\tau^*} dS(\mathbf{y})}_{\text{Monopole}}$$

*cosine directivity*                      *omnidirectional*

The acoustic field is caused by  
imposed velocity + forces exerted on the fluid at the surfaces